higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

NATIONAL CERTIFICATE (VOCATIONAL)

INTRODUCTION TO INFORMATION SYSTEMS
NQF LEVEL 2

(10041002)

13 November 2019 (X-Paper)
09:00–12:00

This question paper consists of 10 pages.

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TIME: 3 HOURS
MARKS: 150

INSTRUCTIONS AND INFORMATION

1. Answer ALL the questions.
2. Read ALL the questions carefully.
3. Number the answers according to the numbering system used in this question paper.
4. Start each SECTION on a NEW page.
5. Use only blue or black ink.
6. Write neatly and legibly.

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SECTION A

QUESTION 1

1.1 Various options are given as possible answers to the following questions. Choose the answer and write only the letter (A–D) next to the question number (1.1.1–1.1.10) in the ANSWER BOOK.

1.1.1 An organisation has people working … toward a common goal.

A apart
B together
C hard
D overtime

1.1.2 … are people elected by the company to oversee the running of the organisation.

A Staff
B Security
C Management
D Merchandisers

1.1.3 When an information system is being developed, the designer must consider … as a factor to take into account.

A system maintenance
B making money
C overtime
D climate

1.1.4 Computers affect management in … by providing reliable and accurate information.

A online purchases
B gaming
C blackouts
D decision making

1.1.5 … property refers to intangible property as a result of creativity.

A Physical
B Owners
C Sole
D Intellectual

1.1.6 In early computing the first computer used was in the form of an …

A abacus.
B calculator.
C automatic machine.
D typewriter.

1.1.7 The … key will remove characters to the right of the cursor.

A backspace
B delete
C esc
D enter

1.1.8 A … is used to create an electronic image of a hardcopy document.

A stylus
B mouse
C scanner
D keyboard

1.1.9 … is used to turn on the numeric keypad.

A Alt
B F1
C Caps lock
D Num lock

1.1.10 A/an … is used to cut large sheets of paper to make banners or posters.

A inkjet printer
B laser printer
C cutter plotter
D pen plotter

(10 × 1) (10)

1.2 Complete the following sentences by using the words in the table below. Write only the missing word next to the question number (1.2.1–1.2.10) in the ANSWER BOOK.

antivirus	organising	technology	electricity	arrow
binary	strategies	procedures	input	free

1.2.1 The … function ensures that a manager is prepared for the day-to-day running of the organisation.

1.2.2 An information system should support the goals and … of the company.

1.2.3 All information systems require … to guide users.

1.2.4 Some companies offer … versions of their antivirus software that come with limitations.

1.2.5 … software scans can be run manually or automatically depending on the settings.

1.2.6 An electromechanical computing device is one that uses … to power mechanical parts.

1.2.7 Data is represented by … numbers on a computer.

1.2.8 A/An … device is any device that can be used to provide input into a computer.

1.2.9 Prolonged use of computers may lead to … overload.

1.2.10 The … keys can be used to move the cursor on the screen.

(10 × 1) (10)

1.3 Indicate whether the following statements are TRUE or FALSE. Choose the answer and write only 'True' or 'False' next to the question number (1.3.1–1.3.10) in the ANSWER BOOK.

1.3.1 Being able to understand people is a characteristic of a good decision maker.

1.3.2 A manager should be systematic when analysing information.

1.3.3 Information systems aim to replace the human workforce with machines.

1.3.4 A worm will replicate itself without the user's knowledge.

1.3.5 Computers play a vital role in the healthcare profession.

1.3.6 Touch screens were used as input in 3rd-generation computers.

1.3.7 Special-purpose computers can be modified by the user to perform alternative tasks.

1.3.8 Ergonomic keyboards are shaped to relieve strain on the user.

1.3.9 The primary function of a terminal is to allow the user to input data.

1.3.10 The system unit is made up of a monitor, mouse and keyboard.

(10 × 1) (10)

1.4 Choose a word or term from COLUMN B that matches a description from COLUMN A. Write only the letter (A–J) next to the question number (1.4.1–1.4.10) in the ANSWER BOOK.

COLUMN A		COLUMN B	
1.4.1	Looking at a situation from various points of view	A	information system
		B	risk
1.4.2	Interrelated information technologies that work together to store, process, manipulate and distribute data securely	C	system files
		D	firewall
1.4.3	A centralised storage facility where information can be found	E	arithmetic logic unit
		F	SRAM
1.4.4	A possibility that a threat will cause loss or damage	G	SD card
1.4.5	A protection mechanism that monitors incoming and outgoing network traffic	H	critical thinking
		I	resolution
1.4.6	A type of random access memory that is used for caching	J	knowledge base
1.4.7	A type of mobile storage device used in digital cameras		
1.4.8	The number of pixels a monitor can show at a given time		
1.4.9	The part of the CPU that performs logical operations		
1.4.10	A type of file that contains the instructions the computer needs to run programs		

(10 × 1) (10)

1.5 Give ONE word or term for each of the following acronyms. Write only the word or term next to the question number (1.5.1–1.5.10) in the ANSWER BOOK.

1.5.1 DSS

1.5.2 GM

1.5.3 FAQs

1.5.4 UPS

1.5.5 MIS

1.5.6 ROM

1.5.7 PC

1.5.8 CPU

1.5.9 DRAM

1.5.10 KB

(10 × 1) (10)
[50]

TOTAL SECTION A: 50

SECTION B

QUESTION 2

2.1 Describe an *organisation*. (2)

2.2 All organisations have support functions that assist the company in performing its core functions.

Suggest TWO departments that would be necessary to perform support functions in a retail organisation. (2 × 2) (4)

2.3 Describe why a manager must plan adequately. (2 × 2) (4)

2.4 Give THREE examples of decisions lower management would make in a retail organisation. (2 × 2) (6)

2.5 You are the operational manager at Asti Manufacturers. A machine operator informs you that one of the machines has malfunctioned. You need to investigate the problem.

Discuss in sequence the first THREE decision making steps that you would need to take. (3 × 2) (6)
[22]

QUESTION 3

3.1 Explain the relationship between *data* and *information*. (4)

3.2 Suggest TWO uses of a knowledge base. (2 × 2) (4)

3.3 Give TWO reasons why companies should be using information systems. (2 × 2) (4)

3.4 You are required to design an information system for Asti Manufacturers.

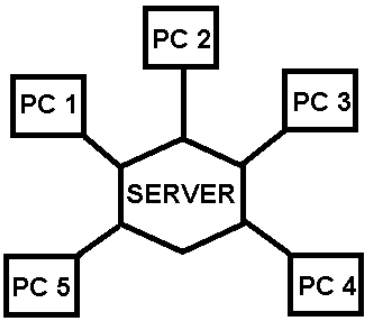
3.4.1 Create a list consisting of FIVE basic information system elements your design will need. (5 × 2) (10)

3.4.2 State ONE factor to consider when designing an information system. (1)

[23]

QUESTION 4

4.1 Study the diagram below and answer the questions.



4.1.1 Identify the type of information system depicted in the diagram.

4.1.2 Suggest a type of configuration to be used when departments have their own servers. (2 × 2) (4)

4.2 Recommend a type of management information system that will be used to prepare financial reports. (2)

4.3 Differentiate between *mandatory access control* and *discretionary access control*. (4)

4.4 Name the TWO categories of internal threats. (2 × 2) (4)

4.5 By using an example, describe how ransomware works. (3 × 2) (6)

4.6 State THREE ways computers could assist students in TVET. (3 × 2) (6)

4.7 Name TWO types of malware. (2 × 2) (4)

[30]

QUESTION 5

5.1 Smartphones are powerful devices. They can be used to make calls and receive text messages as well as emails.

List THREE other tasks that smartphones are capable of carrying out. (3 × 2) (6)

5.2 Describe the concept of *artificial intelligence*. (2 × 2) (4)

5.3 Give TWO examples of convergence technologies. (2 × 2) (4)

5.4 5.4.1 Differentiate between *input* and *output* peripheral devices. (2 × 2) (4)

5.4.2 Name the built-in input device that is used to control the cursor on a laptop. (1)

5.5 5.5.1 Suggest a processing mode that enables multiple users to process tasks at the same time. (2)

5.5.2 Give a reason for your answer in QUESTION 5.5.1 above. (2)

[23]

QUESTION 6

6.1 Describe how a stored program works. (3 × 2) (6)

6.2 List THREE advantages of using a computer system. (3 × 2) (6)

6.3 Explain the purpose of a system unit. (2)

6.4 Name the function key used for help in Windows. (2)

6.5 Differentiate between *primary* and *secondary storage*. (2 × 2) (4)

6.6 Asti Manufacturers urgently require a fax machine as well as a printer for their office. They do not have enough space for two separate machines. As a consultant you are called upon to advise management.

6.6.1 Propose a solution to deal with their need for a fax machine and a printer. (2)

6.6.2 Give TWO disadvantages of implementing the solution proposed in QUESTION 6.6.1 above. (4)

6.7 The receptionist at Asti Manufacturers needs to email pictures of a completed product to a client.

The pictures are large and take long to upload.

6.7.1 Suggest a method of making the picture file size smaller so that they upload faster.

6.7.2 Briefly describe how the method mentioned in QUESTION 6.7.1 above works.

(2 × 2) (4)

6.8 Redraw the following table and suggest TWO practical uses for:

CD-RW	DVD-R
1.	1.
2.	2.

(8)

6.9 Keyboard entry and direct entry differ. Keyboards use keystrokes to enter information, and direct-entry devices do not.

6.9.1 Describe a scenario where keyboard entry would be used to capture information.

6.9.2 Describe a scenario where a direct-entry device would be used to capture information.

(2 × 2) (4)

6.10 Using an example, describe how a dumb terminal works. (3)

6.11 Study the hard disk drive below and answer the questions.



6.11.1 State TWO advantages of using a hard disk drive. (4)

6.11.2 State a disadvantage of using the RAID 0 (striping) storage system. (1)

6.11.3 Name the RAID storage system used for creating a mirror image of data. (2)

[52]

TOTAL SECTION B: 150
GRAND TOTAL: 200

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